PHP Retrieving Data

Fetching Results

Definition

`mysqli_fetch_array()` is a function in PHP used to fetch a result row from a MySQL query as an **array**. It retrieves data from a result set returned by `mysqli_query()` and allows access to the columns either by **column name**, **numeric index**, or **both**.

Syntax

```
<?php
    $row = mysqli_fetch_array(result, resulttype);
?>
```

Parameter	Description
result	The result set returned by mysqli_query().
resulttype (*optional*)	Specifies how the array should be returned. **`MYSQLI_ASSOC`** (column name), **`MYSQLI_NUM`** (index-based), **`MYSQLI_BOTH`**

How It Works

- Executes after a successful SELECT query.
- Fetches **one row at a time** from the result set.
- Each call moves the internal pointer to the next row.
- Returns:
    - An array → if a row exists
    - NULL → if no more rows are available

Sample Codes

```
<?php
    $query = mysqli_query($con, "SELECT id, name FROM users");
    while ($row = mysqli_fetch_array($query)) {
        echo $row['name']; // associative
        echo $row[1];      // numeric
    }
?>
```

Counting Rows

Definition

`mysqli_num_rows()` is a PHP function used to **determine the number of rows** in a result set returned by a MySQL query. It is commonly used after executing a `SELECT` query to check how many records were retrieved.

Syntax

```
<?php
    $count = mysqli_num_rows(result)
?>
```

Parameter	Description
result	The result set returned by mysqli_query().

How It Works

- Used after executing a `SELECT` query.
- Counts how many rows are present in the result set.
- Helps in:
    - Checking if data exists
    - Validating queries
    - Conditional logic (e.g., login systems, duplicate checks)

Sample Codes

```
<?php
    $query = mysqli_query($con, "SELECT * FROM users");
    $count = mysqli_num_rows($query);
    echo "Total rows: " . $count;
?>
```

Session

Definition

A **PHP Session** is a way to store information (variables) on the **server** so that it can be used across multiple pages of a website. Unlike cookies, session data is **not stored on the user's browser**, making it more secure for sensitive data. It is stored in a global associative array: `$_SESSION`

How Sessions Work

1. A session is started using **`session_start()`**
2. PHP generates a **unique session ID**
3. The session ID is stored in the user's browser (usually via a cookie)
4. Actual session data is stored on the **server**
5. Every page using the session retrieves data using the same session ID

Sample Codes

```php
<?php
    session_start();
    $_SESSION['username'] = "harold";
    $_SESSION['role'] = "admin";

    echo $_SESSION['username']; // Output: harold
?>
```

Destroying a Session

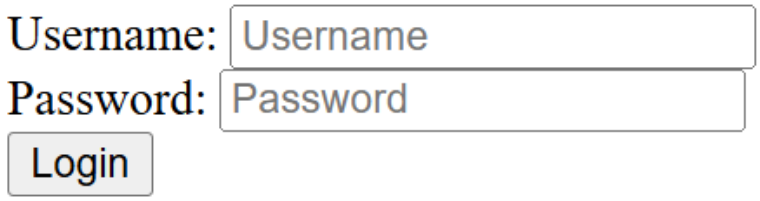
```php
<?php
    session_start();
    session_destroy(); //Destroy all session

    unset($_SESSION['username']); //Unset the username session variable
?>
```

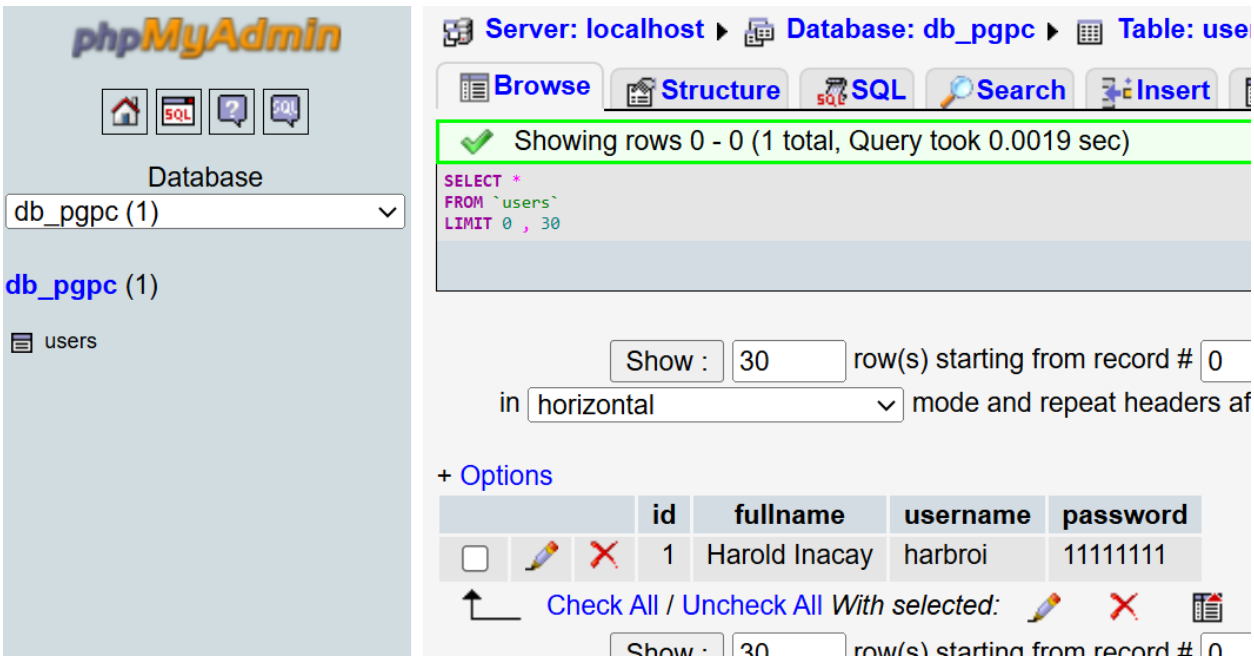
Sample Codes

Login

Design



Database




```
CREATE TABLE IF NOT EXISTS `users` (
  `id` int(11) NOT NULL AUTO_INCREMENT,
  `fullname` text NOT NULL,
  `username` text NOT NULL,
  `password` text NOT NULL,
  PRIMARY KEY (`id`)
)
```

Codes

```
<?php
    $con = mysqli_connect("localhost", "root", "", "db_pgpc");
    session_start();
?>
<html>
    <body>
        <form method="POST">
            Username: <input type="text" name="username" placeholder="Username"
required><br>
            Password: <input type="password" name="password"
placeholder="Password" required><br>
            <input type="submit" name="login" value="Login">
            <?php
                if (isset($_POST['login'])) {
                    $username = $_POST['username'];
                    $password = $_POST['password'];
                    $query = "SELECT * FROM users WHERE username='$username' AND
password='$password'";
                    $result = mysqli_query($con, $query);
                    if (mysqli_num_rows($result) > 0) {
                        $_SESSION['username'] = $username;
                        echo "<script>window.location='main.php';</script>";
                    } else {
                        echo "<br/><span style='color: red;'>Invalid username or
password.</span>";
                    }
                }
            ?>
        </form>
    </body>
</html>
```

List Table Records

Design

Records

ID	Full Name	Username	Password
1	Harold Inacay	harbroi	11111111
2	Maria Santos	mariasantos	11111111

Codes

```
<?php
    $con = mysqli_connect("localhost", "root", "", "db_pgpc");
?>
<html>
    <body>
        <b>Records</b>
        <table border="1" cellspacing="0">
            <tr>
                <th>ID</th>
                <th>Full Name</th>
                <th>Username</th>
                <th>Password</th>
            </tr>
            <?php
            $query = mysqli_query($con, "SELECT * FROM users");
            while($row = mysqli_fetch_array($query)){
            ?>
                <tr>
                    <td><?php echo $row["id"]; ?></td>
                    <td><?php echo $row["fullname"]; ?></td>
                    <td><?php echo $row["username"]; ?></td>
                    <td><?php echo $row["password"]; ?></td>
                </tr>
            <?php
            }
            ?>
        </table>
    </body>
</html>
```